Ramsey/Cass/Koopmans Model Cheatsheet

Macroeconomics I - Lecture 2

UPF, CREi, BSE & CEPR

Notation and Definitions

Aggregate Variables

- K_t: Aggregate capital stock
- L_t: Aggregate labor (population)
- Y_t: Aggregate output
- $A_t \equiv B_t$: Technology level (labor-augmenting, $\dot{A}/A = x$)
- C_t: Aggregate consumption
- N_t: Family size, $N_t = N_0 e^{nt}$ with $N_0 = 1$

Per Worker Variables ($k = K/L$, $y = Y/L$, $c = C/L$)

- $k_t = K_t/L_t$: Capital per worker
- $y_t = Y_t/L_t$: Output per worker
- $c_t = C_t/L_t$: Consumption per worker

Per Effective Worker Variables ($\tilde{k} = K/(AL)$, etc.)

- $\tilde{k}_t = K_t/(A_t L_t)$: Capital per effective worker
- $\tilde{y}_t = Y_t/(A_t L_t)$: Output per effective worker
- $\tilde{c}_t = C_t/(A_t L_t)$: Consumption per effective worker
- Alternative notation: $\hat{k}_t$, $\hat{y}_t$, $\hat{c}_t$ (used in lecture)

Parameters

- n: Population growth rate
- x: Technology (TFP) growth rate
- δ: Depreciation rate
- ρ: Subjective discount rate ($\rho > 0$)
- σ: CRRA curvature parameter ($-\frac{u_{cc}c}{u_c}$)
- s: Savings rate (endogenous in Ramsey, exogenous in Solow)

Model Setup

Environment

- Single homogeneous good (consumed or invested)
- Closed economy
- Exogenous population growth: n
- Exogenous technology growth: x (labor-augmenting)
- Representative household and firm
- Three competitive markets: capital, labor, final good

Firm Problem

The representative firm maximizes profits:

$$\Pi_t = F(K_t, A_t L_t) - (r_t + \delta)K_t - w_t L_t$$

$$F_K(K_t, A_t L_t) = r_t + \delta \quad \text{(marginal product of capital)}$$
$$F_L(K_t, A_t L_t) = w_t \quad \text{(marginal product of labor)}$$

Using per-effective units: $y = f(k)$ where $f(k) = F(k, 1)$

$$f'(\tilde{k}) = r + \delta$$

Household Budget Constraint

Per worker form:

$$c_t + \dot{a}_t + n a_t = w_t + r_t a_t$$

where $a_t = A_t / L_t$ (assets per worker).

Derivation: Household Optimization

Household Problem

Maximize lifetime utility:

$$\max_{\{c_t, a_t\}} \int_0^\infty u(c_t) e^{-(\rho-n)t} \, \mathrm{d}t$$

Subject to:

$$c_t + \dot{a}_t + n a_t \leq w_t + r_t a_t, \quad a_0 \text{ given}$$

No-Ponzi Game: $\lim_{t\to\infty} a_t e^{-\int_0^t (r_s - n)\mathrm{d}s} \geq 0$

Current-Value Hamiltonian Setup

$$\mathcal{H} = u(c_t) + \nu_t[w_t + (r_t - n)a_t - c_t]$$

where ν_t is the co-state (shadow price of assets).

First-Order Conditions **Control variable (c):**

$$\mathcal{H}_c = 0 \Rightarrow \nu_t = u_c(c_t)$$

State variable (a):

$$\mathcal{H}_a = \rho\nu_t - \dot{\nu}_t \Rightarrow \dot{\nu}_t = -(r_t - n)\nu_t$$

State equation:

$$\mathcal{H}_\nu = \dot{a}_t = w_t + (r_t - n)a_t - c_t$$

Euler Equation Derivation

Take logs of $\nu_t = u_c(c_t)$:

$$\log \nu_t = \log u_c(c_t) - (\rho - n)t$$

Differentiate with respect to t:

$$\frac{\dot{\nu}_t}{\nu_t} = \frac{u_{cc}\dot{c}_t}{u_c} - (\rho - n)$$

Substitute $\dot{\nu}_t = -(r_t - n)\nu_t$:

$$-(r_t - n) = \frac{u_{cc}c_t}{u_c}\frac{\dot{c}_t}{c_t} - (\rho - n)$$

Rearrange to obtain:

$$\frac{\dot{c}_t}{c_t} = \left[-\frac{u_{cc}(c_t)c_t}{u_c(c_t)}\right]^{-1}(r_t - \rho) \qquad \text{(Euler Equation (General))}$$

For CRRA utility $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$:

$$\boxed{\frac{\dot{c}_t}{c_t} = \frac{1}{\sigma}(r_t - \rho)} \qquad \text{(Euler Equation (CRRA))}$$

Intuition: Households save more if $r > \rho$ (return exceeds discount rate). Higher σ reduces the response to interest rate changes.

Equilibrium and Dynamic System

Market Clearing

- Labor market: $L_t = N_t$
- Asset market: $K_t = N_t a_t$

Aggregate Resource Constraint

From household BC and market clearing:

$$\dot{k}_t = y_t - c_t - (n + \delta)k_t \qquad \text{(Per worker)}$$

Per-effective-worker form:

$$\dot{\tilde{k}}_t = f(\tilde{k}_t) - \tilde{c}_t - (n + x + \delta)\tilde{k}_t \qquad \text{(Per effective)}$$

Complete Dynamic System

$$\dot{\tilde{k}} = f(\tilde{k}) - \tilde{c} - (n + x + \delta)\tilde{k}$$

$$\frac{\dot{\tilde{c}}}{\tilde{c}} + x = \frac{1}{\sigma}(f'(\tilde{k}) - \delta - \rho)$$

Steady State Analysis

Steady State Conditions

Set growth rates to zero ($\gamma_{\tilde{k}} = 0$, $\gamma_{\tilde{c}} = 0$):

$$\tilde{c}^* = f(\tilde{k}^*) - (n + x + \delta)\tilde{k}^* \qquad (\dot{\tilde{k}} = 0)$$

$$f'(\tilde{k}^*) = \delta + \rho + \sigma x \qquad \text{(Modified Golden Rule)}$$

Cobb-Douglas Case

Let $f(\tilde{k}) = \tilde{k}^\alpha$ with $0 < \alpha < 1$:

$$\tilde{k}^* = \left(\frac{\alpha}{\delta + \rho + \sigma x}\right)^{\frac{1}{1-\alpha}}$$

$$\tilde{y}^* = \left(\frac{\alpha}{\delta + \rho + \sigma x}\right)^{\frac{\alpha}{1-\alpha}}$$

$$\tilde{c}^* = \tilde{y}^* - (n + x + \delta)\tilde{k}^*$$

Golden Rule Comparison

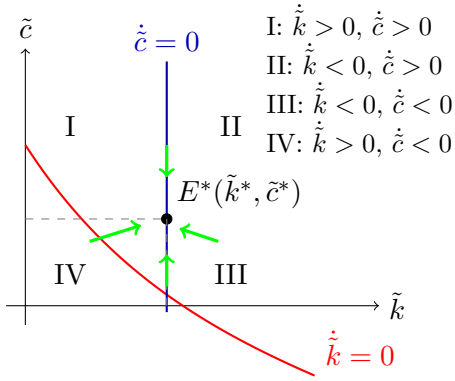
- **Solow Golden Rule**: $f'(\tilde{k}_{GR}) = n + x + \delta$
- **Ramsey Modified Golden Rule**: $f'(\tilde{k}^*) = \delta + \rho + \sigma x$

Since $\rho > 0$ and $\sigma > 0$:

$$f'(\tilde{k}^*) > f'(\tilde{k}_{GR}) \Rightarrow \tilde{k}^* < \tilde{k}_{GR}$$

Phase Diagram and Transitional Dynamics

Phase Diagram Loci



- $\dot{\tilde{c}} = 0$ locus: Vertical line at $\tilde{k}^* = f'^{-1}(\delta + \rho + \sigma x)$
- $\dot{\tilde{k}} = 0$ locus: Inverted U-shape: $\tilde{c} = f(\tilde{k}) - (n + x + \delta)\tilde{k}$
- Intersection: Steady state E^*

Saddle Path Stability

- Only one trajectory leads to steady state (stable arm/saddle path)
- Above saddle path: $\tilde{c}$ too high $\Rightarrow$ capital runs out ($\tilde{k} \to 0$)
- Below saddle path: $\tilde{c}$ too low $\Rightarrow$ over-accumulation violates TVC

Transversality Condition and Dynamic Efficiency

TVC Statement

$$\lim_{t\to\infty}(\nu_t a_t) = 0 \quad \Longleftrightarrow \quad \lim_{t\to\infty} \tilde{c}_t^{\sigma}\tilde{k}_t e^{(n+(1-\sigma)x-\rho)t} = 0$$

Since $\tilde{c}$ and $\tilde{k}$ are bounded in SS:

$$\boxed{n + (1-\sigma)x - \rho < 0} \qquad \text{(TVC Condition)}$$

Implications

1. $\tilde{k}^* < \tilde{k}_{GR}$: Steady state always below Golden Rule
2. No dynamic inefficiency: Cannot have too much capital
3. Interest rate in steady state:
$$r^* = \rho + \sigma x$$

The TVC is crucial for picking the *unique* optimal path. Without it, multiple consumption paths are possible.

Comparative Statics

Parameter Changes in Steady State

Parameter	$\tilde{k}^*$	$\tilde{c}^*$	$\tilde{y}^*$	r^*
$\rho \uparrow$ (more impatient)	$\downarrow$	$\downarrow$	$\downarrow$	$-$
$\sigma \uparrow$ (more risk averse)	$\downarrow$	$\downarrow$	$\downarrow$	$-$
$n \uparrow$ (higher pop growth)	$-$	$-$	$-$	$-$
$x \uparrow$ (faster tech growth)	$\downarrow$	$-$	$-$	$\uparrow$
$\delta \uparrow$ (higher depreciation)	$\downarrow$	$\downarrow$	$\downarrow$	$-$

Key insight: ρ and σ have identical effects on steady state (cannot be distinguished from SS data alone).

Savings Rate in Steady State (Cobb-Douglas)

$$s^* = \alpha \frac{n + x + \delta}{\delta + \rho + \sigma x + \delta}$$

From TVC: $n + x < \rho + \sigma x \Rightarrow s^* < \alpha$ (always below capital share)

Problem-Solving Templates

Template 1: Solve Household Optimization (Hamiltonian Method)

1. Write current-value Hamiltonian: $\mathcal{H} = u(c) + \lambda[w + (r - n)a - c]$
2. FOC wrt control (c): $\partial\mathcal{H}/\partial c = 0 \Rightarrow \lambda = u_c(c)$
3. FOC wrt state (a): $\dot{\lambda} = \rho\lambda - \partial\mathcal{H}/\partial a = (n - r)\lambda$
4. Eliminate λ: Differentiate $\lambda = u_c(c)$ and substitute $\dot{\lambda}$
5. Obtain Euler Equation: $\frac{\dot{c}}{c} = \frac{1}{\sigma}(r - \rho)$

Template 2: Find Steady State

1. Transform to per-effective units: $k = K/(AL)$, $c = C/(AL)$
2. Write dynamic equations: $\dot{k} = f(k) - c - (n + x + \delta)k$, $\frac{\dot{c}}{c} = \frac{1}{\sigma}(f'(k) - \delta - \rho)$
3. Set $\dot{k} = 0$, $\dot{c} = 0$
4. Solve system for k^*, c^*
5. For Cobb-Douglas: $k^* = [\alpha/(\delta + \rho + \sigma x)]^{1/(1-\alpha)}$

Template 3: Phase Diagram Analysis

1. Draw $\dot{c} = 0$ locus: vertical at $k^* = f'^{-1}(\delta + \rho + \sigma x)$
2. Draw $\dot{k} = 0$ locus: $c = f(k) - (n + x + \delta)k$ (inverted U)
3. Identify intersection as steady state
4. Add directional arrows in four regions
5. Draw saddle path (stable arm) through E^*
6. Identify which paths violate feasibility vs. TVC

Template 4: Policy Shock / Parameter Change Analysis

1. Identify which locus shifts:
   - ρ, σ, x changes: shift $\dot{c} = 0$ locus
   - n, x, δ changes: shift $\dot{k} = 0$ locus
2. Determine shift direction (left/right)
3. Find new steady state intersection
4. Analyze transitional dynamics along new saddle path

Common Pitfalls

Confusing ρ (subjective discount) with r (market interest rate)

- **Pitfall 1**: Forgetting technology growth x in the accumulation equation. Always use $n + x + \delta$, not just $n + \delta$.
- **Pitfall 2**: Mixing per-worker vs per-effective variables. Remember: $\tilde{k} = k/A$, not k.
- **Pitfall 3**: Using Solow Golden Rule ($f'(k) = n + \delta$) instead of Modified Golden Rule ($f'(k) = \delta + \rho + \sigma x$).
- **Pitfall 4**: TVC is essential. Without it, multiple steady states possible.
- **Pitfall 5**: In phase diagram, $\dot{c} = 0$ is vertical, $\dot{k} = 0$ is inverted U (not the other way around).
- **Pitfall 6**: The saddle path slopes negatively (downward sloping stable arm).
- **Pitfall 7**: Remember $r^* = \rho + \sigma x$ in steady state, not $r^* = \rho$.

Quick Reference Formulas

Key Equations

$$
\begin{aligned}
&\text{Household BC: } c + \dot{a} + na = w + ra \\
&\text{Euler (CRRA): } \frac{\dot{c}}{c} = \frac{1}{\sigma}(r - \rho) \\
&\text{Capital accumulation: } \dot{k} = y - c - (n + \delta)k \\
&\text{Per-effective: } \dot{\tilde{k}} = f(\tilde{k}) - \tilde{c} - (n + x + \delta)\tilde{k} \\
&\text{Modified Golden Rule: } f'(\tilde{k}^*) = \delta + \rho + \sigma x \\
&\text{TVC: } n + (1 - \sigma)x - \rho < 0 \\
&\text{SS interest rate: } r^* = \rho + \sigma x
\end{aligned}
$$

Cobb-Douglas Steady State

$$
\begin{aligned}
\tilde{k}^* &= \left[\frac{\alpha}{\delta + \rho + \sigma x}\right]^{1/(1-\alpha)} \\
\tilde{y}^* &= \left[\frac{\alpha}{\delta + \rho + \sigma x}\right]^{\alpha/(1-\alpha)} \\
s^* &= \alpha \frac{n + x + \delta}{\delta + \rho + \sigma x + \delta}
\end{aligned}
$$

Comparison: Solow vs Ramsey

	Solow	**Ramsey**
Savings rate s	Exogenous	Endogenous (Euler equation)
Steady state k^*	$s/(\delta + n)$ determined	Modified Golden Rule
Dynamics	Always stable	Saddle path stable
Efficiency	May be inefficient	Always dynamically efficient